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Predicting Commodity Returns Through Image-Based Price Patterns

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ABSTRACT

We examine the predictability of commodity futures returns using image-based price patterns extracted from open-high-low-close (OHLC) charts. Applying convolutional neural networks (CNNs) to US commodity futures data, we extract predictive signals without predefined patterns such as momentum or mean reversion. Empirical results demonstrate that image-based predictions enhance predictive accuracy, particularly over short- and medium-term horizons, with 20-day OHLC images yielding the most robust performance. Compared with traditional financial predictors, CNNs capture nonlinear dependencies while retaining unique explanatory power. Panel regressions confirm that image-based predictions are correlated with established return factors. However, transfer learning—from the US to the Chinese markets—proves ineffective in commodity futures markets, highlighting the necessity of market-specific adaptation.

JEL Classification: G13, G17, G45

1 | Introduction

Traditional empirical studies have focused on well-known phenomena such as momentum effects, mean reversion, and seasonal patterns, providing valuable insights into market dynamics (Poterba and Summers 1988; Jegadeesh 1991; Jegadeesh and Titman 1993). However, these methods often rely on predefined patterns and linear assumptions, which may overlook the complex, evolving nature of financial markets. Recent research has increasingly adopted data-driven methodologies to uncover latent patterns in financial data (Mullainathan and Spiess 2017; Gu et al. 2020; Leippold et al. 2022). Jiang et al. (2023) show that open-high-low-close (OHLC) charts in stock markets contain rich visual information, which when analyzed with convolutional neural networks (CNNs), can effectively predict short- and long-term returns. This suggests that price patterns reflect complex investor behavior and microstructural features that traditional models fail to capture.

Although Jiang et al. (2023) demonstrate the effectiveness of image-based price pattern prediction in equity markets, whether the same approach can capture the external-shock-driven and often regime-shifting dynamics of commodity futures remains an open question. Commodity futures markets exhibit unique characteristics that differentiate them from traditional asset classes such as equities and bonds. These markets are driven primarily by supply-demand dynamics, seasonality, storage costs, geopolitical factors, as well as external shocks such as weather events, natural disasters, and trade policy changes, leading to return patterns that are distinct from those influenced by macroeconomic indicators (Bianchi et al. 2015). Moreover, commodity futures markets are characterized by higher leverage, greater volatility, and a diverse participant base—including producers, consumers, hedgers, and speculators—which amplifies the nonlinear nature of price behavior. Additionally, the low correlation between commodity futures returns and those of equities and bonds underscores their importance for asset allocation and risk management (Heide et al. 2025).

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Existing research on commodity futures markets has largely concentrated on predefined price patterns such as momentum, mean reversion, and seasonality (Miffre and Rallis 2007; Bianchi et al. 2015; Li et al. 2024). While these strategies have shown varying degrees of success, they frequently struggle to adapt to evolving market conditions, presenting limitations in the highly volatile and dynamic environment of commodity futures markets where exogenous shocks can rapidly invalidate fixed pattern assumptions (Wang and Zhang 2024).

To address these challenges, we use CNNs to analyze OHLC price charts in the commodity futures market. Originally developed for image recognition tasks, CNNs are well suited to capturing complex spatial patterns and extracting latent features from structured visual inputs. In recent years, the financial literature has increasingly explored the use of CNNs. However, prior research has largely concentrated on applications involving the prediction of individual asset prices or tasks such as image-based clustering and classification (Obaid and Pukthuanthong 2022), with a focus on equities (Cohen et al. 2020), oil futures (Göncü et al. 2024; Ren et al. 2024), and stock indexes (Chen et al. 2016; Hoseinzade and Haratizadeh 2019). Notably, there has been little attention devoted to large-scale return forecasting at various commodity futures with a comprehensive portfolio analysis and transfer learning between the US and Chinese commodity markets examined in our study.

By explicitly modeling the spatial signatures left by weather events, geopolitical headlines, and inventory shocks on price charts, our study demonstrates that CNNs offer a powerful alternative to different trend-based and machine learning (ML) strategies. We address this gap by providing novel empirical evidence on the application of CNNs in asset pricing and by exploring the learning process of CNNs and their effectiveness in developing systematic trading strategies in the commodity futures market.

Our ML–based framework autonomously identifies and extracts latent patterns with predictive power from a vast data set of price charts, offering a flexible tool for forecasting commodity futures returns. Whereas traditional analysis of these patterns often involves subjective judgment, ML algorithms like CNNs offer a systematic, objective means of identifying and validating these patterns, thereby enabling the timely incorporation of external shocks into forecasting models and enhancing the robustness of trading signals, making them particularly well suited for financial applications.

The findings of this study highlight the effectiveness of CNNs in capturing nonlinear, complex patterns embedded in OHLC charts, significantly improving return prediction accuracy over short- and medium-term horizons. Predictions based on 20-day OHLC charts achieve the best balance between signal stability and predictive performance, consistently outperforming traditional strategies such as momentum and mean reversion in terms of Sharpe ratios (SRs) and risk-adjusted returns. Furthermore, our analysis demonstrates that image-based models extract predictive signals that are strongly correlated with financial factors such as momentum, short-term reversals, and proximity to 52-week highs while autonomously capturing nonlinear interactions and temporal dependencies.

Jiang et al. (2023) report that patterns learned from US stocks are effective in international markets through transfer learning. However, our study finds that image-based models trained on US commodity futures data fail to generalize effectively to Chinese commodity futures. Structural differences across commodity markets, shaped by local supply and demand conditions, regulatory frameworks, and the composition of market participants, constrain the generalizability of predictive patterns. These findings underscore the need for market-specific adaptation and indicate that direct transfer without retraining is ineffective for commodity futures return prediction.

This article contributes to the literature by extending image-based price prediction methods to the commodity futures market. Our approach avoids reliance on predefined assumptions or linear models, instead employing a flexible, data-driven framework to autonomously identify and validate predictive patterns. This makes it particularly suited to the nonlinear and dynamic nature of commodity futures markets. Our study reinforces and extends the findings of Jiang et al. (2023) for US stocks. It offers practical implications by providing tools for developing trading strategies and optimizing asset allocation.

The remainder of this article is organized as follows. Section 2 reviews the literature on commodity futures trading strategies. Section 3 introduces the data, explains OHLC chart construction, and outlines preprocessing steps for CNN implementation. Section 4 presents empirical results, evaluating CNN model performance in return prediction and comparing it to other trading strategies. Section 5 explores CNN interpretability by analyzing learned patterns from OHLC charts and discussing transfer learning across markets. Section 6 concludes.

2 | Literature Review on Trading Strategies in Commodity Futures

Early empirical work interprets expected commodity futures returns through the lens of risk premium theory. In practice, variables such as the basis, the slope of the term structure, and empirical proxies for convenience yield reliably forecast excess returns (Fama and French 1988; Erb and Harvey 2006). The parallel literature sorts contracts on lagged price dynamics. Portfolios formed on monthly momentum, calendar seasonality, and long-horizon reversal deliver sizable risk-adjusted profits, although performance varies across business cycle phases and is sensitive to roll costs (Sørensen 2002; Gorton and Rouwenhorst 2006; Miffre and Rallis 2007). Factor extensions such as the commodity capital asset pricing model (Adrian et al. 2013) and the basis-momentum two-factor model (Szymanowska et al. 2014; Boons and Prado 2019) improve cross-sectional explanatory power, yet they remain linear and largely time invariant. More recent research embeds seasonality and inventory cycles, documenting state-dependent interactions between carry and momentum (Gorton et al. 2013; Cheng et al. 2023).

While these approaches are firmly grounded in economic theory, their parametric structure limits their ability to capture higher-order interactions, multicollinearity, and structural breaks, motivating a shift toward data-driven methods. The recent literature has adopted ML techniques to relax functional-

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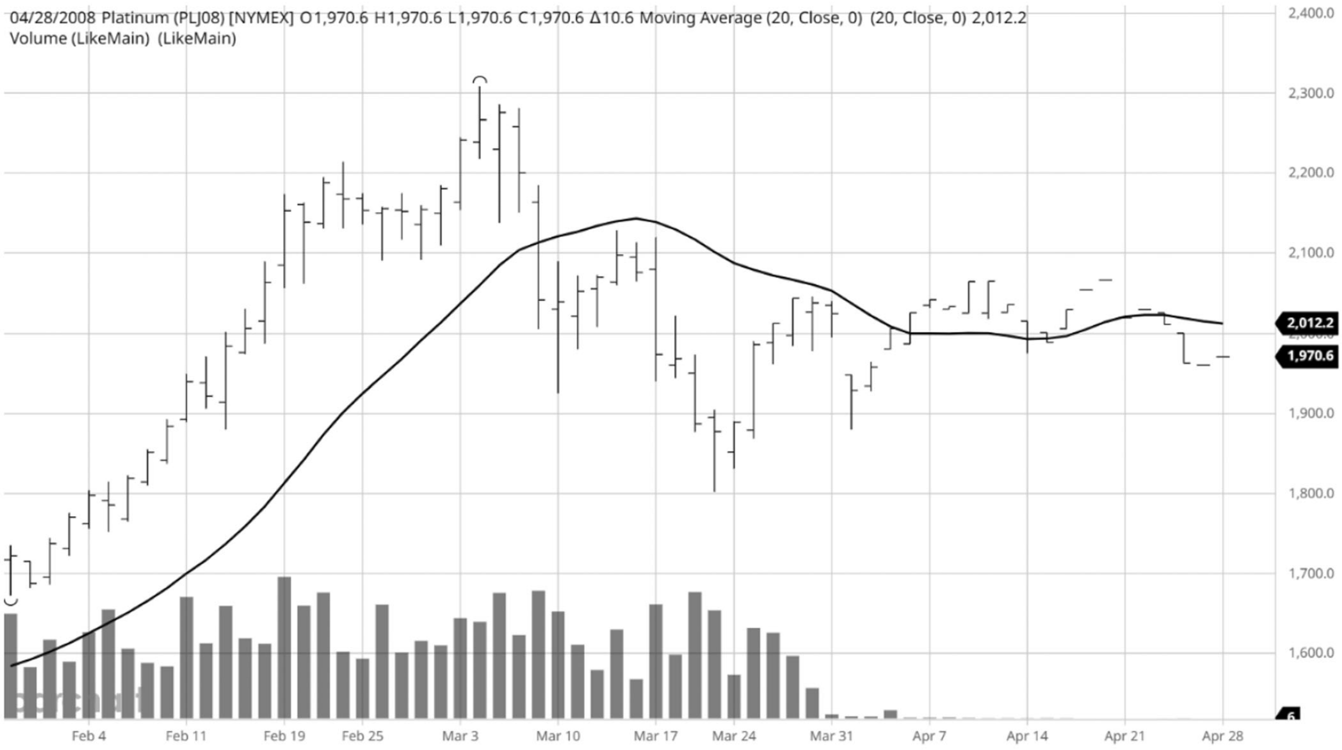

FIGURE 1 | Platinum Apr '08 (PLJ08). This figure presents an OHLC chart sourced from Barchart (https://www.barchart.com/futures/quotes/PLJ08/interactive-chart), depicting Platinum Apr '08 (PLJ08) data. The chart features a 20-day moving average price line and daily trading volume bars, spanning the period from January 29, 2008, to April 30, 2008.

form constraints and exploit richer information sets (Gu et al. 2020). Classical ML algorithms, including support vector machines and gradient-boosted trees, capture nonlinear interactions among carry, momentum, volatility, and macroeconomic variables, leading to significant gains in out-of-sample predictive accuracy (Gong et al. 2022; Wang and Zhang 2024). Han and Kong (2022) apply adaptive Lasso to uncover serial dependence in futures returns and document significant predictability.

Deep networks push the frontier further. Rad et al. (2023) employ neural networks to study the links between 128 macro financial predictors and subsequent futures returns, showing that economic-state variables contain information about commodity risk premia. Herrera et al. (2019) compare neural networks and random forests with traditional econometric models over longer horizons and find persistent outperformance in energy contracts. Ren et al. (2024) and Göncü et al. (2024) predicted the oil futures price with a CNN model.

Despite these advances, previous studies still focus on a limited set of contracts or extend linear factor models only marginally. Consequently, both the return-predictive power of image-based representations and their robustness across a broad cross-section of commodities remain largely unexplored. Addressing this gap, we develop an image-based framework that learns directly from OHLC price images for a large panel of liquid futures. Benchmarking against canonical factor strategies and state-of-the-art ML algorithms reveals that the CNN captures additional information and materially enhances risk-adjusted performance.[1]

3 | Data and Methodology

3.1 | OHLC

Many popular websites, such as Bloomberg, Yahoo Finance, and Google Finance, provide historical price charts for various financial assets. Figure 1 illustrates an example of Platinum Apr '08 (PLJ08) data displayed on Barchart in a standard price chart format. The chart includes OHLC bar graphs depicting the daily open, high, low, and close prices, along with a 20-day moving average of closing prices. Daily trading volume is displayed at the bottom of the chart.

Although such charts are readily available online, we generate our own price charts from scratch to enable controlled experimentation by adjusting the amount of information visible to the CNN model. Our charts adhere to the basic format of Figure 1, using black OHLC bar graphs where the high and low prices correspond to the top and bottom of the central vertical bar, while the open and close prices are marked by small horizontal lines on the left and right sides, respectively. Each day occupies a three-pixel-wide space: one pixel for the central bar, one for the open marker, and one for the close marker.

Following Jiang et al. (2023), our commodity futures images consist of OHLC bar graphs spanning 5, 20, or 60 continuous days, corresponding approximately to weekly, monthly, and quarterly price trajectories. Thus, an n-day image has a width of $3n$ pixels. Instead of raw prices, we use adjusted returns, converting the open, close, high, and low prices into relative values. After assigning dates, we fix the image height

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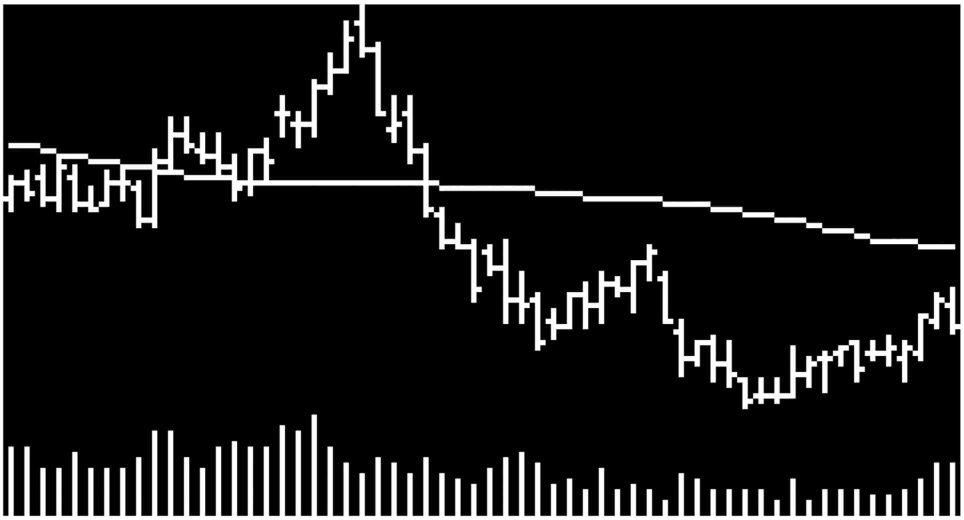

FIGURE 2 | Sixty-day OHLC image with volume bar and moving average. This figure depicts a 60-day commodity futures contracts image, including volume bars and the 60-day moving average line.

and scale the vertical axis so that the maximum and minimum of the OHLC path align with the top and bottom of the image. As a result, all images for a given period share identical pixel dimensions.

To optimize storage efficiency, we use a black background with white for all visible chart elements. Since black pixels are represented by (0,0,0), this design minimizes data storage requirements by producing sparse images. In line with Bloomberg and similar platforms, we omit color distinctions for "up" and "down" days, as price direction is already encoded in the open and close markers. This allows us to focus on two-dimensional (2D) pixel matrices without tracking red, green, and blue pixel intensities.

Additional information, such as trading volume and moving averages, is incorporated into the OHLC graphs. Each image includes a moving average line (e.g., a 20-day moving average for a 20-day window) and a volume bar chart. The volume chart occupies the bottom one-fifth of the image, while the OHLC plot fills the top four-fifths. To maintain consistency, we scale the maximum volume within an image to match the upper limit of the volume section, adjusting the other volume bars proportionally.

Figure 2 provides an example from our data set, featuring a 60-day OHLC chart for corn futures starting on August 9, 2007. This chart encapsulates key details about price trends, volatility, intraday and overnight return patterns, and trading volume. The design balances informational richness and storage efficiency, ensuring a comprehensive data set for the CNN model.

3.2 | Commodity Futures

We utilize the same diverse range of markets as Keloharju et al. (2016), covering 20 commodity futures and 4 metal forward contracts across agricultural products, energy, livestock, and metals. Using daily closing prices of near-month futures contracts traded on exchanges such as the Chicago Board of Trade (CBT), the Intercontinental Exchange (ICE), New York Mercantile Exchange (NYMEX), Chicago Mercantile Exchange (CME), and the London Metal Exchange (LME), we compute 5-, 20-, and 60-day logarithmic returns for these 24 futures and forward contracts. The sample period spans from January 1980 to October 2024, incorporating the most recent data from our provider, Commodity Systems Inc. For contracts introduced after 1980, we begin the analysis from their first trading day. Table 1 presents the sample details and summary statistics for 20-day log commodity returns.

Commodity futures pose unique challenges due to rollovers as contracts approach expiration and the existence of multiple contracts with different expiration dates for the same product. Many existing studies address this issue by excluding contracts with fewer than 15 trading days to expiration and immediately rolling into the next contract (Miffre and Rallis 2007; Shen et al. 2007) or by gradually rolling into the next contract (Wang and Yu 2004; Marshall et al. 2008). However, a key limitation of these approaches is that multiple contracts for the same product often exhibit significant differences in liquidity.

To prevent information leakage and maintain data validity, we select the futures contract with the highest trading volume over the past 5 trading days as the next-day contract, provided it has more than 15 trading days until expiration (Han et al. 2016; Yang et al. 2018). This approach assumes that the most liquid contract is likely to remain the most active on the following trading day. When a rollover occurs, we adjust the price series by rebasing the price using the true daily returns of the new contract.

For visualization, we use the newly constructed continuous price series and aggregated trading volumes to generate OHLC charts, rendering them in grayscale pixels. In cases of trading halts or market suspensions, we do not leave gaps in the charts. Instead, to maintain visual continuity, we backfill the OHLC chart using the most recent N valid trading days, ensuring reliable price pattern representation.

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TABLE 1 | Summary statistics of 20-day returns in US commodity futures market.

Sector	Commodity	Exchange	Start date	Obs	Avg	SD	Min	p25	p75	Max
Agriculture	Corn	CBT	1980/01/01	11,275	−0.0041	0.0662	−0.3716	−0.0400	0.0320	0.4195
	Wheat	CBT	1980/01/01	11,275	−0.0063	0.0718	−0.3521	−0.0506	0.0361	0.5077
	Soybeans	CBT	1980/01/01	11,275	−0.0002	0.0605	−0.3200	−0.0358	0.0337	0.2633
	Kansas City HRW Wheat	CBT	1980/01/01	11,276	−0.0033	0.0691	−0.3312	−0.0432	0.0333	0.4521
	Cocoa	ICE-US	1980/01/01	11,223	−0.0029	0.0794	−0.4384	−0.0528	0.0456	0.4869
	Coffee	ICE-US	1980/01/01	11,222	−0.0045	0.0942	−0.4811	−0.0617	0.0460	0.6033
	Cotton	ICE-US	1980/01/01	11,233	−0.0011	0.0676	−0.3426	−0.0415	0.0384	0.3530
	Sugar	ICE-US	1980/01/01	11,223	−0.0038	0.0971	−0.3870	−0.0621	0.0530	0.5405
Energy	Crude Oil	NYMEX	1983/04/08	10,414	0.0016	0.0996	−0.9544	−0.0491	0.0602	0.6687
	NY Harbor ULSD	NYMEX	1980/01/01	11,234	0.0019	0.0876	−0.5657	−0.0482	0.0548	0.4977
	Gasoline-Reformulated Blendstock	NYMEX	1984/12/11	9991	0.0058	0.1031	−1.2023	−0.0479	0.0656	0.6010
	Henry Hub Natural Gas	NYMEX	1990/04/11	8654	−0.0172	0.1315	−0.6005	−0.0994	0.0670	0.5462
	Brent Crude	ICE-EU	1980/01/01	9263	0.0050	0.0934	−0.8130	−0.0461	0.0623	0.4605
	Low-Sulphur Gasoil	ICE-EU	1981/04/14	11,082	0.0039	0.0894	−0.6455	−0.0452	0.0556	0.5161
Livestock	Feeder Cattle	CME	1980/01/01	11,282	0.0003	0.0408	−0.2855	−0.0228	0.0263	0.1788
	Live Cattle	CME	1988/07/01	11,282	0.0012	0.0397	−0.2869	−0.0211	0.0258	0.1759
	Lean Hogs	CME	1980/01/01	11,282	−0.0022	0.0717	−0.4833	−0.0453	0.0441	0.3752
Metals	Aluminum	LME	1987/06/12	9434	0.0010	0.0563	−0.2862	−0.0337	0.0360	0.2835
	Nickel	LME	1980/01/01	11,303	0.0014	0.0891	−0.5978	−0.0502	0.0517	0.7508
	Lead	LME	1980/01/01	11,311	0.0018	0.0696	−0.4687	−0.0367	0.0419	0.3432
	Zinc	LME	1988/09/02	9124	0.0017	0.0684	−0.4568	−0.0369	0.0439	0.2772
	Gold	COMEX	1980/01/01	11,247	−0.0010	0.0478	−0.3237	−0.0267	0.0255	0.3665
	Silver	COMEX	1980/01/01	11,247	−0.0044	0.0891	−1.1838	−0.0470	0.0427	0.4206
	Copper	COMEX	1980/01/01	11,247	0.0019	0.0706	−0.5443	−0.0385	0.0433	0.3928

Note: This table presents the descriptive statistics for 24 underlying assets, categorized into four sectors: Agriculture, Energy, Livestock, and Metals. The data set includes 20 commodity futures and 4 metal forward contracts. *Obs* represents the number of trading days. The columns Avg, SD, Min, p25, p75, and Max report the average, standard deviation, minimum, 25th percentile, 75th percentile, and maximum values of log returns over a 20-day period, respectively. All futures begin on January 1, 1980, or the first available trading day for each asset, and extend through October 2024.

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Our price trend analysis constructs continuous return series by adjusting for contract rollovers. For each image, we standardize the first day's closing price to one and calculate window-level returns, RET_{t+q}, $q = 5$, 20, 60, using subsequent daily closing prices. This ensures that returns are comparable across different contracts and consistent across analysis windows. In commodity futures, rolling from one contract to another may introduce artificial price discontinuities. We rebase the price series at each rollover point using actual returns to preserve continuity.

$$RET_{t+q} = \ln(p_{t+q}) - \ln(p_t) \tag{1}$$

We analyze market data images spanning the past 5, 20, or 60 days, assigning binary labels of 1 or 0 based on whether the returns over the subsequent 5, 20, or 60 days are positive or nonpositive. As a result, our primary analysis consists of nine separately estimated models. Following Gu et al. (2020) and acknowledging the stochastic nature of CNN optimization, we independently retrain the CNN five times for each model configuration and average the predictions to enhance robustness.

3.3 | Training the CNN Models

We follow Jiang et al. (2023) closely in implementing the CNN models. The models impose cross-parameter constraints, significantly reducing parameterization, which makes them easier to train and more effective for predictions using relatively small data sets (Krizhevsky et al. 2017; Obaid and Pukthuanthong 2022). Additionally, the CNN integrates various tools that enable the model to accommodate deformations and relocations of key objects within the images. The CNN transforms raw images into a set of predictive features through a series of stacked operations, ultimately generating predictions.

The CNN consists of three primary operations: convolution, activation, and pooling. Convolution operates similarly to kernel smoothing. It scans the image both horizontally and vertically, summarizing the content of neighboring regions for each element in the image matrix (Krizhevsky et al. 2017). Activation applies a nonlinear transformation, specifically the "leaky ReLU" to the output of the convolution filter on an element-wise basis. The final operation in the building block is "max pooling," which scans the input matrix and returns the maximum value from neighboring regions in the image. This reduces the data's dimensionality and noise. The final CNN layer is fully connected and activated by a softmax function.

The objective of the CNN is to predict binary outcomes, where a value of 1 indicates a positive return within the specified forward-looking period, and a value of 0 indicates otherwise. Consequently, the fitted value of the CNN represents an estimate of the probability of a positive outcome.

Each image in our data set is represented as a pixel value matrix, where black or white pixels are assigned values of 0 or 255, respectively, and each element corresponds to the grayscale intensity at the corresponding position in the image. Standard image-processing CNNs, referred to as 2D CNNs, move their rectangular filters horizontally and vertically across the image matrix. Traditional price prediction methods rely on statistical modeling of historical time-series data or one-dimensional (1D) models. In contrast, our approach leverages 2D CNNs, as representing the data as images enables convolutional filters to capture nonlinear spatial correlations between various price patterns. Images integrate information about directional price movements, deviations from moving average trends, price volatility, and trading volume into a unified representation.

For instance, considering price direction and volatility simultaneously in traditional time-series models would require restrictive assumptions and manual feature engineering. Incorporating volatility information typically necessitates nonlinear transformations of the price series using stochastic volatility or generalized autoregressive conditional heteroskedasticity models. However, 2D CNN eliminates the need for manual feature design by autonomously extracting predictive patterns from the market data series. Additionally, 2D CNN applied to price-chart images addresses the limitations of 1D models (e.g., long short-term memory [LSTM]), which are constrained to analyzing linear relationships. For example, a 3×2 filter can treat "no change," "minor increase," and "major increase" as distinct features, assigning different weights to these patterns for final predictions. When applied to images, these filters function as indicator mechanisms for detecting different price movements.

We divide the entire data set into training, validation, and test sets. For model estimation and validation, we used data from January 1980 to December 2000 as the training sample. Within this data set, 70% images are randomly selected for training, while the remaining 30% are used for validation. Random sampling for training and validation helps balance the distribution of positive and negative labels in the classification task, mitigating potential biases caused by prolonged bullish or bearish market trends. The remaining years (from January 2001 to October 2024) of data are reserved as the out-of-sample test data set.

The task of predicting future returns is inherently a binary classification problem. The training process minimizes the standard objective function for classification problems, which is the cross-entropy loss function:

$$L(y, \hat{y}) = -y \ \log(\hat{y}) - (1 - y)\log(1 - \hat{y}) \tag{2}$$

where $\hat{y}$ is the softmax output from the CNN in the final step. If the predicted probability matches the label perfectly ($\hat{y} = y$), the loss is 0; otherwise, the loss is positive.

The model consists of two to four convolutional layers, each followed by batch normalization (BatchNorm) to mitigate internal covariate shifts during training (Ioffe and Szegedy 2015). Leaky ReLU (Maas et al. 2013) is used as the activation function for the convolutional layers, with the slope parameter set to 0.01 to prevent the "dead neuron" issue often associated with traditional ReLU activations. Each convolutional operation is immediately followed by max pooling, with a pooling window size of 2×1. This progressively reduces the size of feature maps, lowers computational complexity, and enhances the model's focus on critical features.

The convolutional layer parameters dynamically adapt to the length of the time window (e.g., the span of the input data). For example, with a 5-day window input, the model includes two

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convolutional layers with dilation and stride parameters set to 1×1.[2] This adaptive design enables the model to extract features from data spanning different time horizons.

After the convolutional layers, the resulting feature maps are flattened into 1D vectors and passed to fully connected layers. The number of input features in the fully connected layers is determined dynamically by the output of the convolutional layers. For instance, assuming specific input tensors and convolutional operations, the flattened feature dimensions are 256 for a 5-day window, 512 for a 20-day window, and 1024 for a 60-day window. The fully connected layers map these features to the number of target classes (e.g., for binary or multiclass classification tasks).[3] Finally, the output of the CNN is a set of contract-level estimates for commodities, indicating the probability of a positive subsequent return over short (5-day), medium (20-day), and long (60-day) horizons.[4]

To prevent overfitting, an early stopping mechanism is employed. Specifically, training stops if the validation loss does not improve for two consecutive epochs. Additionally, whenever the validation loss decreases, the model saves its current weights, ensuring that the best-performing weights are used for final deployment.

4 | CNN Prediction for US Commodity Futures Returns

4.1 | Portfolio Construction

Rebalancing is conducted on the first day of standard weekly, monthly, and quarterly cycles. The probabilities derived from the image data up to and including the trading day before the rebalancing day are sorted into tercile portfolios based on out-of-sample CNN estimates for the probability of a positive subsequent return. Portfolios maintain proportional weights for the assets in each tercile. For futures at the boundary of terciles, their allocations are split proportionally between the relevant portfolios.

At the time of rebalancing, the opening prices are used to calculate the differences between the current holdings and the target portfolio weights. Adjustments are made through buying and selling, while ensuring that at least 0.1% of the total portfolio value is held as cash. We also construct a long-short portfolio ("H-L"), where the long positions correspond to the highest tercile ("High") and the short positions correspond to the lowest tercile ("Low"). To address the inconsistency between the holding period of each portfolio and the predictive horizon of the model and to ensure uniformity in future return periods across different time frames for more robust pattern predictions, the predicted values derived from images are aligned to a standardized horizon of 5-, 20-, or 60-day intervals.

We conduct evaluations on a weekly, monthly, or quarterly basis. The first day of each cycle is designated as the rebalancing date, minimizing deviations and accounting for realistic market frictions in both experimental and practical settings. The notation "Ix/Ry" represents models that use x-day image predictions to forecast returns over the subsequent y-day holding period. For each portfolio, the total asset value, which includes both the market value of holdings and the cash balance, is updated at the end of each trading day. Daily returns (R_d) and their variance (σ_d) are then calculated based on these updates, and the Sharpe Ratio (SR) is computed using annualized daily returns.

$$SR = \frac{R_d}{\sigma_d} \times \sqrt{days_{perYear}} \tag{3}$$

4.2 | Portfolio Performance

Image-based return prediction represents a technical price trend signal. In the stock market, Jiang et al. (2023) demonstrate that image-based strategies are most effective over relatively short horizons (5 days). Table 2 presents the prediction results of CNN models employing image-based strategies in the US commodity futures market. Panel A reports the results for short-horizon (5-day-ahead) returns, Panel B focuses on medium-horizon (20-day-ahead) returns. Panel C highlights long-horizon (60-day-ahead) returns, corresponding to weekly, monthly, and quarterly horizons, respectively. The table evaluates the predictive power of these strategies from an economic perspective by summarizing the performance of portfolios ranked based on CNN predictions. We emphasize the annualized average portfolio return and the annualized SR of the holding period returns.

Each panel reports equal-weighted tercile portfolios. The low tercile represents futures with the lowest probability of positive future returns and consistently achieves highly negative SRs, exceeding −1.0 for all image sizes. The SR increases monotonically across the upward terciles when predicting returns using 20-day images.

To benchmark these results, we also report the performance of alternative holding-period strategies based on MOM (2-12 momentum), STR (short-term reversal), WSTR (1-week short-term reversal), and TREND (the trend strategy of Han et al. 2016) signals.[5] In short-term predictions, image-based strategies generally achieve higher SRs than traditional price-trend-based strategies (except WSTR). In medium- and long-horizon predictions, image-based strategies exhibit similar results. Within the image-based strategies, using 20-day images to predict future returns consistently outperforms predictions made using 5- or 60-day images.

Specifically, for the I20/R20 horizon, the CNN model yields the highest average return (5.66%) and SR (0.38) for the "H-L" portfolio, followed by TREND (4.49% return and 0.22 SR from the CNN), while the other strategies exhibit negative returns. Similarly, for the I20/R60 horizon, the CNN model achieves a higher return (2.55%) and SR (0.16) compared with other strategies. The results remain consistent when considering the High portfolio for the I20/R20 horizon (7.13% return and 0.42 SR from the CNN) and the I20/R60 horizon (2.55% return and 0.16 SR from the CNN).

To evaluate portfolio stability, Table 2 reports the average turnover rate, which is calculated using the standard formula for fund turnover. Specifically, for each rebalancing period, the turnover rate is determined by dividing the absolute value of total purchases ($|Buy|_t$) and total sales ($|Sell|_t$) by the

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TABLE 2 | Portfolio performance of the CNN model with other strategies.

	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel A: Short-horizon (one-week) portfolio performance														
Low	5.92	0.38	3.53	0.22	1.96	0.13	4.11	0.23	3.24	0.18	1.49	0.08	4.46	0.23
2	0.09	0.01	4.05	0.25	5.33	0.32	2.98	0.20	5.26	0.35	4.67	0.31	3.89	0.26
High	6.93	0.41	5.11	0.31	5.56	0.34	5.32	0.27	3.98	0.21	6.92	0.37	4.11	0.23
H-L	1.01	0.07	1.59	0.11	3.60	0.24	1.21	0.06	0.75	0.03	5.42	0.26	−0.35	−0.02
Turnover (%)	3180.69		3101.49		3013.85		646.76		1633.79		3170.02		1525.92	
	I5/R20		**I20/R20**		**I60/R20**		**MOM/R20**		**STR/R20**		**WSTR/R20**		**TREND/R20**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel B: Middle-horizon (one-month) portfolio performance														
Low	4.06	0.26	1.47	0.09	4.49	0.28	5.41	0.30	4.68	0.26	4.93	0.27	0.59	0.03
2	4.29	0.26	4.07	0.25	1.24	0.08	2.46	0.16	7.00	0.47	3.99	0.26	6.73	0.44
High	3.79	0.23	7.13	0.42	6.55	0.38	4.76	0.26	0.36	0.02	3.43	0.19	5.08	0.28
H-L	−0.26	−0.02	5.66	0.38	2.05	0.13	−0.65	−0.03	−4.31	−0.21	−1.51	−0.08	4.49	0.22
Turnover (%)	748.57		759.03		697.94		264.96		753.08		766.71		548.15	
	I5/R60		**I20/R60**		**I60/R60**		**MOM/R60**		**STR/R60**		**WSTR/R60**		**TREND/R60**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel C: Long-horizon (one-quarter) portfolio performance														
Low	4.90	0.31	3.14	0.21	3.97	0.27	5.64	0.32	3.75	0.20	4.63	0.26	4.48	0.24
2	2.62	0.16	4.71	0.30	3.15	0.19	3.58	0.24	4.40	0.30	2.67	0.17	3.43	0.23
High	5.32	0.31	5.69	0.32	5.88	0.33	2.72	0.14	4.91	0.26	4.98	0.28	4.90	0.26
H-L	0.42	0.03	2.55	0.16	1.91	0.12	−2.92	−0.14	1.15	0.06	0.36	0.02	0.42	0.02
Turnover (%)	242.42		243.22		250.71		146.22		265.55		252.35		229.93	

Note: This table compares the portfolio performance of the CNN model with other strategies in the US commodity futures market. Panels A–C present results for short-horizon (one-week), middle-horizon (one-month), and long-horizon (one-quarter) performance, respectively. "Ret" is expressed as a percentage (%). Following Jiang et al. (2023), we benchmark the CNN strategy against four traditional price trend strategies: 2–12 month momentum (MOM), 1-month short-term reversal (STR), 1-week short-term reversal (WSTR), and the trend strategy (TREND) proposed by Han et al. (2016), which incorporates short-, medium-, and long-term price trends.

portfolio's total equity value (*Equity*$_t$). Following Pástor et al. (2017, 2020), the absolute values of total purchases and sales are computed based on the hypothetical trades generated by the portfolio's rebalancing process under the proposed strategy. The analysis assumes an initial portfolio value and updates holdings over time according to the strategy-determined weights, without considering transaction costs. The average turnover rate (*Turnover*) is then annualized to reflect the portfolio's trading activity over the years:

$$Turnover = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{1}{T_i} \sum_{t=1}^{T_i} \frac{|Buy|_t + |Sell|_t}{Equity_t} \right) \tag{4}$$

where T_i is the number of trading periods within year i, and N is the total number of years over which the turnover is averaged.

We use the opening price of each period as the rebalancing price to ensure meaningful turnover calculations. All signals are generated using data available at the previous trading day's close, eliminating future information leakage and allowing sufficient time for rebalancing preparation. Additionally, all performance metrics are calculated based on actual daily returns from the real return curve, ensuring that observed volatility accurately reflects real-world market fluctuations.

This measure provides a standardized approach to comparing portfolio trading activity across different time periods and strategies. Unlike single-event turnover rates, it aggregates turnover over all trading periods, ensuring consistency in evaluation. Annualization accounts for variations in trading frequency across years, enabling more intuitive comparisons. By averaging normalized turnover rates over multiple years, the reported value reflects long-term trading behavior while mitigating the impact of short-term fluctuations.

To further strengthen the benchmark comparison, we additionally incorporate multifactor strategies from Bianchi et al. (2015), including the standard medium-term momentum strategy (Mom_{12-1}) and the combined momentum-reversal strategies (Mom_{12}–$\text{Ctr}_{\{T\}}$, where $T \in \{18, 24, 36, 48, 60\}$).[6] These strategies integrate medium-term momentum signals with long-term reversal effects and are among the most widely used benchmarks in the commodity futures literature. To ensure comparability with the CNN 2D results reported in Table 2, we use the same test period from 2001 to 2024.

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TABLE 3 | Portfolio performance of different momentum and reversal strategies in commodity futures.

	$\mathbf{Mom_{12-1}}$	$\mathbf{Mom_{12}–Ctr_{18}}$	$\mathbf{Mom_{12}–Ctr_{24}}$	$\mathbf{Mom_{12}–Ctr_{36}}$	$\mathbf{Mom_{12}–Ctr_{48}}$	$\mathbf{Mom_{12}–Ctr_{60}}$
Ann. Ret (%)	−1.65	0.28	0.28	2.10	−1.83	2.02
Ann. Sharpe	−0.079	0.011	0.011	0.086	−0.077	0.083
Ann. volatility	0.207	0.250	0.249	0.244	0.237	0.243
Skewness	0.106	0.104	0.133	0.129	0.091	0.324
Kurtosis	0.939	0.487	0.646	0.901	0.490	0.822
% of positive months	50.37	50.38	49.62	50.40	49.58	45.54
Max drawdown	−0.831	−0.829	−0.793	−0.784	−0.826	−0.797

Note: In this table, we report the long-short portfolio performance of various combinations of momentum and reversal strategies as proposed in Bianchi et al. (2015). To ensure comparability with our CNN 2D results, we use the same test sample period from 2001 to 2024. The strategies include Mom_{12-1}, which is a pure momentum strategy with a 12-month ranking period and a 1-month holding period, as well as momentum-contrarian strategies ($Mom_{12}–Ctr_{\{T\}}$, where $T = 18$, 24, 36, 48, and 60), which apply a double-sort procedure: first ranking based on 12-month momentum and then applying a contrarian ranking over T months, all with a 1-month holding period. We report annualized return in percentage terms (Ann. Ret), annualized Sharpe ratio (Ann. Sharpe), annualized return volatility (Ann. Volatility), skewness, kurtosis, the percentage of positive months, and maximum drawdown.

The findings in Table 3 show that the single-sort momentum strategy (Mom_{12-1}) yields an annualized return of approximately −1.65% with an SR of −0.08. Among the combined momentum-reversal double-sort strategies, the highest annualized return is 2.10% ($Mom_{12}–Ctr_{36}$), while the lowest is −1.83% ($Mom_{12}–Ctr_{48}$). Notably, all these strategies underperform compared with the CNN 2D results in the "Ix/R20" configuration, providing further evidence of the superior predictive performance of the CNN approach.

In Appendix A, we present the predictions of the univariate model at different horizons using the same methodology as in Table 2. As a benchmark, we compare the CNN 1D model with the LSTM models, with results shown in Tables A1 and A2. The performance of the benchmark models consistently falls short of the CNN 2D model across all prediction horizons, highlighting the superior predictive power of the CNN 2D model in forecasting returns using image-based data.

Panels A–C of Figure 3 display the cumulative return curves for short-term, medium-term, and long-term strategies, respectively. Columns (1–3) correspond to strategies that predict returns over the next 5, 20, and 60 days. Each plot illustrates the cumulative return curves for different tercile portfolios, along with the "H-L" portfolio, with the maximum drawdown points highlighted.

Figure 3 presents the cumulative returns of image-based long-short portfolios across different forecasting horizons, highlighting distinct patterns in predictive efficacy. Portfolios 1, 2, and 3 trace the cumulative returns of the bottom-, middle-, and top-tercile portfolios, respectively, whereas the "H-L" portfolio shows the cumulative return of a zero-cost strategy that is long the top tercile and short the bottom tercile. The results indicate that CNN models perform best over medium-term horizons, whereas their effectiveness declines for short- and long-term predictions due to heightened volatility and weaker signal strength.

In Panel B, the medium-term (20-day) predictions deliver the strongest and most stable returns, particularly in I20/R20, where the "H-L" portfolio achieves the highest cumulative return (195.79%) with the lowest drawdown (30.80%). This suggests that CNN models capture market dynamics most effectively at this horizon, balancing predictive accuracy and risk exposure. In contrast, Panel A shows the short-term (5-day) predictions, which exhibit high volatility and inconsistent performance, with the "H-L" portfolio in I5/R5 generating near-zero cumulative returns (−1.75%) and a substantial drawdown (48.91%), indicating that short-term fluctuations are difficult to predict reliably. For long-term (60-day) forecasts in Panel C, cumulative returns decline steadily, with I60/R20 and I60/R60 producing only modest gains (21.16% and 18.00%, respectively) alongside elevated drawdowns (70.83% and 50.00%, respectively). This reflects the model's diminishing predictive power over extended horizons.

Overall, the results confirm that image-based predictions are most effective in medium-term trading strategies, particularly in 20-day horizons, where predictive strength remains robust and risk is better controlled. Short-term predictions suffer from excessive volatility, while long-term forecasts weaken due to signal decay, reinforcing the importance of selecting an optimal predicting window.

5 | Exploring the Learning Process of CNNs

To better understand the patterns identified by image-based models, we employ two complementary approaches. First, we examine the relationship between CNN predictions and a set of widely studied financial signals related to price, risk, and liquidity. Second, we utilize regression-based approximations to analyze the underlying data representations used by CNNs in image-based forecasting.

5.1 | Relationship With Traditional Predictive Factors

Our analysis focuses on several key metrics commonly associated with asset return predictability, including price trends and volatility. Table 4 reports the univariate correlations between

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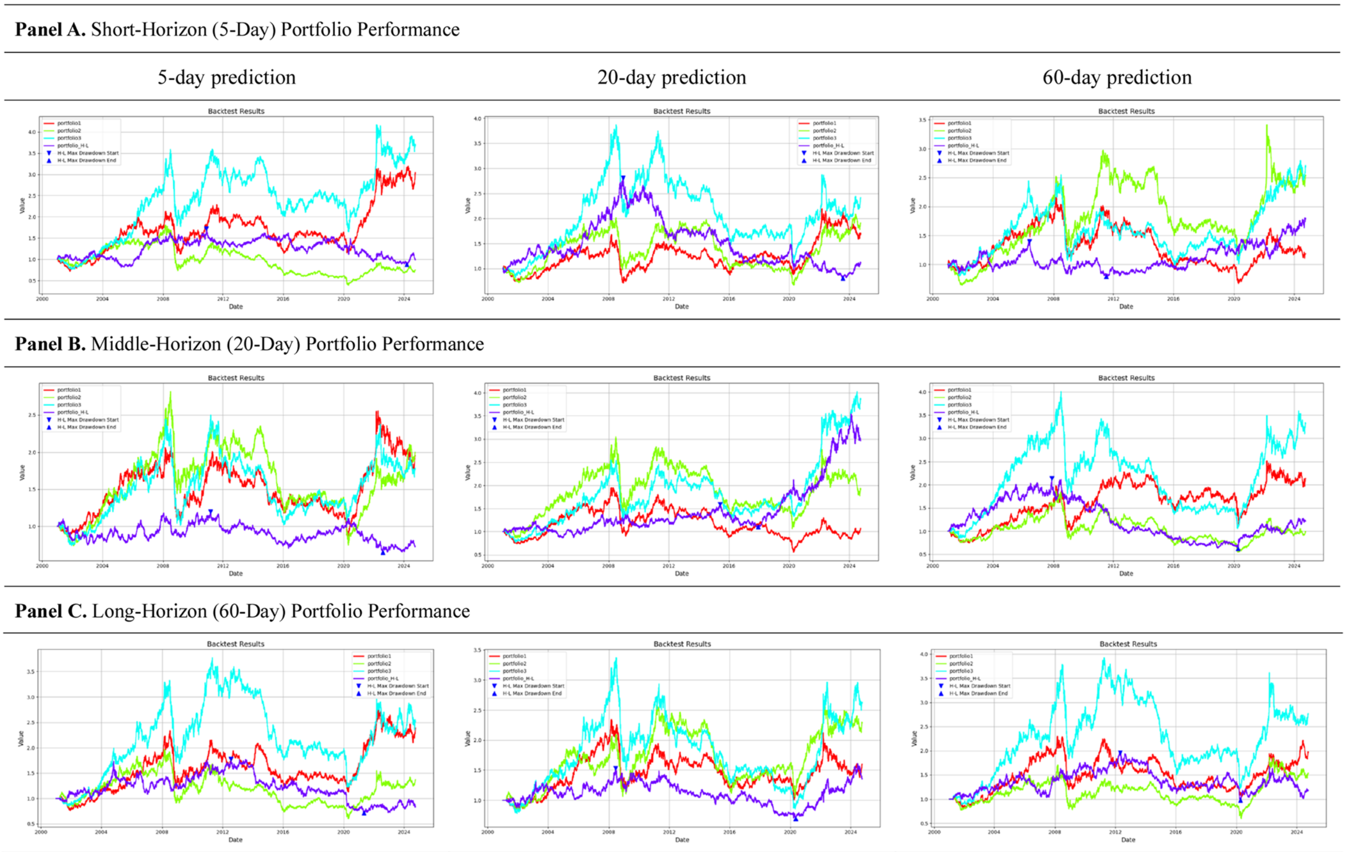

FIGURE 3 | Different horizons portfolio performance of image-based model prediction. This figure presents the cumulative return performance of long-short portfolios based on CNN model predictions across different horizons. Each panel shows the return trends of tercile portfolios ranked by predicted returns. Portfolios 1, 2, and 3 trace the cumulative returns of the bottom, middle, and top tercile portfolios, respectively, while the high-minus-low (H-L) portfolio shows the cumulative return of a zero-cost strategy that is long the top tercile and short the bottom tercile. Results indicate that image-based predictions perform best in the middle horizons, where the H-L portfolio maintains consistent positive returns with lower drawdowns. As the forecasting horizon extends, the return spread narrows, with the 60-day horizon showing weaker profitability, suggesting diminished predictive power over longer periods. These findings underscore the model's effectiveness in capturing short- and middle-horizon price patterns in commodity futures markets while highlighting its limitations for longer-horizon predictions. Panel A. Short-Horizon (5-Day) Portfolio Performance. Panel B. Middle-Horizon (20-Day) Portfolio Performance. Panel C. Long-Horizon (60-Day) Portfolio Performance.

TABLE 4 | Correlation between CNN predictions and future characteristics.

Model	MOM	STR	WSTR	TREND	52WH	Volatility
I5/R5	0.02	−0.12	−0.15	0.00	−0.07	0.00
I5/R20	0.00	−0.17	−0.05	0.00	−0.06	−0.01
I5/R60	0.10	−0.12	0.00	−0.01	−0.08	−0.01
I20/R5	0.09	−0.20	−0.19	0.03	−0.09	−0.02
I20/R20	0.05	−0.16	−0.12	−0.01	−0.07	0.00
I20/R60	0.07	−0.20	−0.06	−0.02	−0.15	0.00
I60/R5	0.09	−0.26	−0.32	0.09	−0.11	−0.01
I60/R20	0.04	−0.21	−0.11	−0.01	−0.12	0.01
I60/R60	0.09	−0.19	−0.08	−0.04	−0.13	0.00

Note: This table presents the average cross-sectional correlation between CNN model predictions and various future characteristic factors (MOM, STR, WSTR, TREND, 52WH, and Volatility) over the test sample. Among these factors, STR exhibits the strongest association with CNN predictions, ranging from 12% to 26%. WSTR follows a similar pattern but weakens as the prediction horizon extends from 5- to 60-day images, aligning with the CNN capturing a weekly STR pattern. Consistent with Jiang et al.'s (2023) findings in the stock market, MOM shows its highest correlation (9%) with long-horizon forecasts from large images (I60/R60) and its lowest correlation (2%) with short-horizon forecasts from small images (I5/R5).

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these characteristics and the predictions generated by the image-based models. Specifically, we compute cross-sectional rank correlations between each factor and the CNN forecasts on a period-by-period basis, then report the time-series average of these cross-sectional correlations over the test sample period.

Among these factors, STR exhibits the strongest association with CNN model predictions, with correlation coefficients ranging from 12% to 26% across different forecasting horizons. WSTR follows a similar pattern, demonstrating a strong correlation with CNN forecasts in shorter horizons but diminishing in strength as the forecasting window extends from 5 to 60 days. This pattern is consistent with the CNN model capturing weekly STR signals.

MOM displays its highest correlation (9%) with long-horizon forecasts (I60/R60) but exhibits a substantially weaker relationship (2%) for short-term predictions (I5/R5). Other predictive factors, including Trend, 52-Week High (52WH), and Volatility, exhibit the strongest correlations with CNN forecasts at the longest horizon (I60), suggesting that image-based models extract information relevant to longer-term return patterns in the commodity futures market. These findings confirm that CNNs can identify meaningful predictive signals from data abstracted into image representations.

To evaluate the joint explanatory power of CNN forecasts and traditional financial characteristics, we employ panel logistic regressions to model the probability of a positive return over 5, 20, or 60 days. The regression model is

$$Logit\left(\Pr\left(CNN_{i,t}^{(m,n)} = 1\right)\right) = \beta_0 + \sum_{k=1}^{K} \beta_k X_{k,i,t} + \epsilon_{i,t} \qquad (5)$$

where $CNN_{i,t}^{(m,n)} = 1$ denotes the CNN-based prediction for commodity i at time t, generated using image representations of the past m trading days to forecast the positive return over the next n days (and 0 otherwise). $X_{k,i,t}$ are cross-section ranked predictors (including MOM, STR, WSTR, TREND, 52WH, Volatility), and $\epsilon_{i,t}$ is the error term.

Table 5 presents the results of panel logistic regressions evaluating the joint explanatory power of image-based return forecasts and standard financial characteristics across different forecast horizons. The dependent variable is a binary indicator of whether a positive return is realized over the subsequent 5, 20, or 60 days, and all regressions are estimated exclusively on the test sample. Panels A, B, and C correspond to prediction horizons of 5, 20, and 60 days, respectively, allowing for a systematic comparison of predictive performance over varying time frames. All predictor variables are transformed into cross-sectional ranks to mitigate distributional distortions and ensure robustness.

TABLE 5 | CNN predictions and standard futures characteristics.

	5D/5P	20D/5P	60D/5P
Panel A: 5-day prediction			
MOM12–1	0.01	0.01	0.05
STR	−0.11***	−0.25***	−0.08
WSTR	−0.21***	0.01	0.1
TREND	−0.02	−0.03	−0.04
52WH	−0.01	−0.03	−0.20*
Volatility	0.02	0.06	0.04
McFadden R^2R^2	−0.45	−0.21	−0.32
	5D/20P	**20D/20P**	**60D/20P**
Panel B: 20-day prediction			
MOM12–1	0	0.06	−0.16*
STR	−0.12*	−0.26***	−0.24***
WSTR	−0.25***	−0.10**	0.09
TREND	0.03	−0.02	−0.06
52WH	−0.21**	−0.11*	−0.58***
Volatility	−0.02	0.07	0.21***
McFadden R^2	0.72	0.88	2.10
	5D/60P	**20D/60P**	**60D/60P**
Panel C: 60-day prediction			
MOM12–1	0.06	−0.05	−0.17*
STR	−0.14	−0.42***	−0.18*
WSTR	−0.43***	−0.01	0.08
TREND	0.11	−0.01	−0.15*
52WH	−0.22*	−0.18*	−0.78***
Volatility	0.18*	0.08	0.31***
McFadden R^2	1.29	1.22	2.98

Note: This table presents the joint explanatory power of all features for image-based predictions, with Panels A, B, and C corresponding to forecasts for 5-, 20-, and 60-day horizons, respectively. We estimate a panel logistic regression using the cross-sectional ranks of all factors. In the multivariate regression, STR and WSTR emerge as the most significant explanatory variables across image sizes, while only 52WH is significant for 60-day images. This suggests that while the CNN detects signals associated with well-known predictive variables, its image-based predictions retain distinct and unique variation. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively, based on Newey–West standard errors.

In the 5-day prediction horizon (Panel A), STR and WSTR emerge as the most significant predictors, both exhibiting strong negative associations with future returns. STR has a coefficient of −0.11 in 5D/5P (significant at the 1% level), reinforcing the presence of short-term mean reversion in price movements. Similarly, WSTR shows a coefficient of −0.21 in 5D/5P (significant at the 1% level), further supporting the importance of short-term price corrections. The statistical significance of these variables persists in the 20-day predictive horizon but weakens as the forecast window extends to 60 days, suggesting that the mean-reverting nature of short-term returns dissipates over longer periods.

In contrast, the 20-day prediction horizon (Panel B) reveals a more persistent role of STR and WSTR, with coefficients of −0.12 (significant at the 10% level) and −0.25 (significant at the 1% level) in 5D/20P, respectively. The coefficient of WSTR in 20D/20P is −0.10 (significant at the 5% level), maintaining statistical significance beyond the shortest forecast window. Additionally, the variable 52WH exhibits stronger predictive power at this horizon, with a coefficient of −0.21 in 5D/20P (significant at the 5% level) and −0.11 in 20D/20P (significant at the 5% level), suggesting that assets trading near their 52-week highs tend to experience downward corrections over intermediate horizons. Volatility becomes positively

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associated with future returns in 20D/20P (coefficient = 0.07) and 60D/20P (coefficient = 0.21, significant at the 1% level), indicating a risk-return trade-off that strengthens over time.

At the 60-day prediction horizon (Panel C), most traditional predictors lose significance except for 52WH, which becomes the dominant explanatory variable. The coefficient of 52WH is −0.20 in 60D/5P (significant at the 5% level) and −0.78 in 60D/60P (significant at the 1% level), demonstrating a consistent downward price correction effect for assets trading near their 52-week highs. This pattern aligns with long-term mean reversion theories, suggesting that commodity futures experiencing prolonged uptrends are more susceptible to reversals over extended periods. Volatility also exhibits increased significance at this horizon, with a coefficient of 0.31 in 60D/60P (significant at the 1% level), reinforcing the notion that riskier assets exhibit stronger return predictability at longer horizons.

The McFadden R^2 values provide additional insights into the overall explanatory power of the models. In Panel A, negative McFadden R^2 (ranging from −0.45 to −0.21) indicate that traditional financial predictors alone provide limited explanatory power, underscoring the distinctiveness of image-based forecasts. However, in the 20- and 60-day horizons, McFadden R^2 values increase substantially (e.g., 0.72 in 5D/20P, 2.98 in 60D/60P), reflecting a greater alignment between standard financial characteristics and return dynamics over longer periods. These patterns suggest that, whereas short-term reversals dominate the predictive landscape in the immediate term, longer-term price movements exhibit stronger associations with structural financial characteristics such as prior highs and volatility.

The findings underscore the time-varying nature of return predictability, where short-term forecasts are dominated by rapid reversals, while long-term predictions increasingly reflect structural price corrections and risk-related factors. The diminishing role of STR and WSTR at longer horizons suggests that mean reversion effects are more relevant for short-term trading strategies, whereas investors with longer holding periods may benefit from incorporating measures such as prior highs and volatility. Moreover, the improvement in model fit over time indicates that image-based forecasts capture unique predictive information that complements traditional financial predictors, particularly in short-term price movements where conventional factors exhibit weaker explanatory power.

Table 6 reports the results of panel logistic regressions examining the predictive power of image-based return forecasts for future returns, controlling for standard return predictors. The regressions are estimated exclusively on the test sample to prevent data leakage, maintaining the independence of CNN parameters from the data set used for evaluation. The general form of the regression equation can be expressed as:

$$Logit(\Pr(R_{i,t,t+n} > 0)) = \beta_0 + \beta_1 CNN_{i,t}^{(m,n)} + \sum_{k=2}^{K} \beta_k X_{k,i,t} + \epsilon_{i,t} \tag{6}$$

where the dependent variable equals 1 if the realized n-day ($R_{i,t,t+n}$) is positive on commodity i starting from day t (and 0 otherwise). The definitions of $CNN_{i,t}^{(m,n)}$ and $X_{k,i,t}$ are the same as in Equation (5).

Three regression models are estimated for each CNN specification: (i) a univariate regression using only CNN forecasts, (ii) a regression including traditional characteristics but excluding CNN forecasts, and (iii) a multivariate regression incorporating both CNN forecasts and standard financial predictors. Since the independent variables and the dependent variable remain consistent across the CNN specifications, the coefficients for financial predictors remain unchanged across different models. The results suggest that the CNN forecasts retain strong predictive power even in the presence of conventional financial predictors, indicating that deep learning-based forecasts provide additional insights that are not fully captured by traditional return characteristics.

The image-based return forecasts exhibit strong and statistically significant predictive power across all return horizons. In Panel A, the estimated coefficients on CNN increase from 0.23 at the shortest horizon (I5/R5) to 0.72 at the longest horizon (I60/R5), all significant at the 1% level. Panel B shows that CNN achieves the best performance in predicting 20-day returns with a coefficient of 1.04 at I60/R20. Incorporating standard return predictors increases the coefficient to 1.07, suggesting a slight improvement in predictive power. This increasing trend suggests that CNN models effectively extract return-relevant signals that persist and strengthen over time, particularly in medium- and long-term forecasts. The stability of CNN's predictive power across the univariate and multivariate regressions implies that its forecasts provide incremental information that is not merely a transformation of traditional financial predictors.

Among conventional financial predictors, past-week return (WSTR) exhibits a statistically significant positive relationship with future returns, as reflected in their coefficients of 0.10 (I5/R5, significant at the 1% level) and 0.19 (I60/R5). This finding aligns with the well-documented short-term momentum effect, where recent gains tend to persist over short to medium horizons. Conversely, 52WH consistently exhibits a strong negative association with future returns, with coefficients ranging from −0.26 (I5/R5, significant at the 1% level) to −0.19 (I60/R5). This pattern is consistent with mean-reversion effects, where futures trading near their historical highs tend to experience downward price corrections over time.

In Panel B's 20-day return predictions, 52WH shows a significant negative predictive effect, with its significance decreasing as the prediction horizon lengthens. Other traditional predictors, such as momentum (MOM) and short-term trend indicators (TREND), display weaker and less consistent relationships with future returns, reinforcing the limitations of linear financial predictors in fully capturing complex return patterns. For instance, MOM remains statistically insignificant across all specifications. In Panel C, however, all standard return predictors lose their predictive power in long-term forecasts, further emphasizing the challenges of using traditional predictors for long-term market prediction.

The out-of-sample McFadden R^2 metrics further validate the CNN's superiority. Positive values at intermediate horizons (e.g., 1.72 for I20/R5 or 0.53 for I60/R60) indicate that the model

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TABLE 6 | CNN, future returns, and standard future characteristics.

	I5/R5			I20/R5			I60/R5		
Panel A: 5-day prediction									
CNN	0.23***		0.23***	0.46***		0.44***	0.72***		0.71***
MOM		−0.05	−0.05		−0.01	−0.01		0	−0.02
STR		−0.01	0		−0.07	0.01		−0.01	0.06
WSTR		0.10***	0.13***		0.15*	0.13		0.19	0.16
TREND		0.02	0.02		0.08	0.08		0.21*	0.2
52WH		−0.26***	−0.25***		−0.2	−0.16		−0.19	−0.15
Volatility		0.07**	0.07**		−0.07	−0.07		−0.23	−0.25*
OOS McFadden R^2	−0.02	0.32	0.1	−0.34	1.72	−0.2	−3.65	1.55	−3.57
	I5/R20			**I20/R20**			**I60/R20**		
Panel B: 20-day prediction									
CNN	0.16**		0.15*	0.54***		0.51***	1.04***		1.07***
MOM		0.05	0.04		0.12*	0.09		−0.02	0.04
STR		0.04	0.06		−0.05	−0.01		0.15	0.16
WSTR		0.08	0.1		−0.04	0.01		−0.06	−0.06
TREND		0	0		−0.01	−0.01		−0.01	−0.05
52WH		−0.34**	−0.34**		−0.18**	−0.15*		−0.26*	0.05
Volatility		0.06	0.06		0.08	0.07		0.11	0.02
OOS McFadden R^2	0.3	−0.4	−0.06	−0.52	−0.37	−0.6	−0.58	−0.31	−0.96
	I5/R60			**I20/R60**			**I60/R60**		
Panel C: 60-day prediction									
CNN	0.12		0.09	0.78***		0.79***	0.65***		0.64***
MOM		0.21	0.21		0.05	−0.02		0.06	0.06
STR		0.03	0.03		−0.05	0.08		−0.05	−0.04
WSTR		−0.01	0.01		−0.03	0.03		−0.03	−0.02
TREND		−0.04	−0.04		−0.02	−0.03		0.09	0.08
52WH		−0.17	−0.17		−0.17	−0.16		−0.17	−0.03
Volatility		−0.1	−0.1		−0.12	−0.16		−0.12	−0.20*
OOS McFadden R^2	0.11	−0.31	−0.17	0.38	−0.25	0.59	0.57	−0.32	0.53

Note: This table presents slope coefficients from panel logistic regressions of future returns on image-based CNN model forecasts and standard future characteristics. The regressions are estimated during the test sample using CNN models trained on the training sample. Additionally, the table reports out-of-sample McFadden R^2 values, comparing the cross-entropy loss of the trained model against a benchmark model that uses the future-level in-sample mean as the predictor. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively, based on Newey–West standard errors.

generalizes effectively to unseen data, capturing persistent market anomalies. In contrast, negative R^2 values at extreme horizons (e.g., −3.57 for I60/R5 or −0.96 for I60/R20) may reflect unmodeled nonlinearities or regime shifts in low-frequency regimes, where logistic frameworks struggle to approximate abrupt market transitions. These findings resonate with the adaptive market hypothesis, suggesting that the CNN's advantage lies in identifying slow-decaying inefficiencies, while short-term noise remains challenging to systematize.

These findings reinforce the superiority of deep learning techniques in identifying complex return signals that extend beyond those captured by traditional financial predictors. The statistically significant and increasing coefficients of CNN forecasts confirm that these models successfully extract persistent return patterns. Meanwhile, the mixed performance of traditional predictors highlights the difficulty of using linear features to capture return dynamics. The strong out-of-sample predictive performance of CNN forecasts further reinforces their applicability in financial markets, suggesting that deep learning–based models can serve as valuable tools for return forecasting in both short-term and long-term investment strategies.

In Appendix B, we employ a logistic regression to approximate the predictive performance of the CNN model. Given that the CNN architecture functions as a probabilistic classifier, logistic regression serves as an appropriate approximation model, enabling a linearized interpretation of the CNN's decision-making process. By estimating the relationships between CNN forecasts and fundamental market variables, we assess the extent to

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TABLE 7 | Portfolio performance of the CNN model for different sectors.

	Agriculture		Energy		Livestock		Metals	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**	**Ret**	**SR**
Low	1.36	0.10	−3.81	−0.11	−3.93	−0.16	6.28	0.44
2	−1.76	−0.13	9.53	0.28	−0.20	−0.01	2.15	0.17
High	6.73	0.32	5.37	0.16	3.35	0.19	9.62	0.41
H-L	5.37	0.26	9.18	0.36	7.28	0.27	3.34	0.16
Turnover (%)	662.67		765.28		635.17		614.88	

Note: This table presents the portfolio performance of the CNN model across different sectors (Agriculture, Energy, Livestock, and Metals, as shown in Table 1). We report the results based on the I20/R20 methodology for its best performance. For individual commodity futures, we are unable to construct separate portfolios, so we utilize the full-sample model, trained across all sectors, to backtest the futures of each sector. The Energy sector yields the highest long-short return at 9.18%, with an SR of 0.36. The Livestock sector follows with a long-short return of 7.28% and an SR of 0.27. The Agriculture and Metals sectors rank third and fourth, with long-short returns of 5.37% and 3.34% and SRs of 0.26 and 0.16, respectively.

which the deep learning model captures return-relevant information beyond traditional financial indicators.

To further validate the predictive performance of our CNN model, we use the full-sample model trained across four sectors (Energy, Livestock, Agriculture, and Metals) to backtest the futures of each sector. As shown in Table 7, the Energy sector generates the highest long-short return at 9.18%, with an SR of 0.36. The Livestock sector follows with a long-short return of 7.28% and an SR of 0.27. The Agriculture and Metals sectors rank third and fourth, with long-short returns of 5.37% and 3.34% and SRs of 0.26 and 0.16, respectively. These results demonstrate the robustness of the CNN model across different market sectors, highlighting its effectiveness in predicting sector-specific futures.

Building on a growing body of research in commodity futures forecasting, recent studies have found that ML methods, particularly deep learning models, are better suited than traditional statistical approaches for capturing nonlinear dynamics and structural shifts in price behavior. For instance, in agricultural markets, Ouyang et al. (2019) employ a long- and short-term time-series network to capture both high- and low-frequency features of multivariate series, achieving significant improvements over benchmark models. Similarly, recent works have introduced text-based and hybrid feature extraction methods, such as the integration of emotional signals or topic modeling from news, to enhance forecasting performance for specific commodities like soybean, pork, and corn (An et al. 2023; Wu et al. 2024; Wang and Liu 2025). While these methods have demonstrated promising accuracy within targeted domains, they are largely designed and evaluated within single-commodity contexts.

In energy and metals markets, prior research has frequently focused on volatility forecasting or directional prediction using neural networks, random forests, or ensemble learning methods (Herrera et al. 2019; Lu et al. 2022; Foroutan and Lahmiri 2024). More recent developments have also introduced feature selection and model interpretability techniques, including the use of Shapley value decomposition to assess the marginal impact of macroeconomic and market-specific variables on return prediction (El Hokayem et al. 2024; Wang and Zhang 2024). Other contributions emphasize the use of high-dimensional data inputs, including text-derived sentiment indices and deep feature representations from models such as transformers, to improve short-term commodity return forecasting (Gong et al. 2022; Gao et al. 2025). However, these studies often require extensive domain-specific calibration or focus on a narrow set of assets, limiting their extensibility.

Our study adopts a unified CNN architecture to forecast futures returns across energy, livestock, agriculture, and metals sectors. The model achieves consistent predictive accuracy and generates robust long-short signals across commodities with diverse fundamentals. Compared with prior methods that often target individual assets or require extensive domain-specific calibration, our approach demonstrates strong generalization across heterogeneous markets. These results contribute to the existing literature and highlight the practical value of deep learning models in commodity return forecasting and portfolio-level applications, particularly in enhancing risk-adjusted performance.

Overall, the results confirm that image-based forecasts extract meaningful return signals that extend beyond conventional financial variables. The logistic regression analysis demonstrates that CNN models systematically incorporate extreme price movements, trend deviations, and liquidity conditions into return expectations. The significant improvement in return classification, even after controlling for traditional predictors, underscores the value of deep learning techniques in financial forecasting, particularly in markets where complex, nonlinear interactions drive price movements.

5.2 | Transfer Learning

Transfer learning enables the application of predictive patterns from one market to another, leveraging insights from data-rich environments to enhance forecasting in markets with limited historical data. Compared with the United States, the Chinese commodity futures market has a shorter history and a more constrained data set, making it challenging to train complex deep learning models effectively. If return-predictive patterns from US commodity futures generalize across markets, their direct application to China could mitigate data limitations and improve forecasting accuracy. However, this generalizability remains an empirical question given the structural differences between the two markets.

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To test the feasibility of international transfer learning, we apply a CNN model trained on US commodity futures to predict returns in the Chinese market. The transfer learning approach directly employs CNN models estimated on US data to forecast Chinese commodity futures returns without retraining. We construct tercile-ranked portfolios based on these forecasts and evaluate their out-of-sample returns and SRs, comparing them to CNN models trained exclusively on Chinese market data.

The data set consists of 36 Chinese commodity futures contracts from 1998 to 2024 (Bianchi et al. 2021; Liu et al. 2021; Ming et al. 2023). The sample is divided into a training period (1998–2008), with 70% allocated for training and 30% for validation, while the post-2009 period serves as the out-of-sample data set. The data set covers contracts from the Shanghai Futures Exchange, Dalian Commodity Exchange, and Zhengzhou Commodity Exchange, spanning four primary sectors: (i) metals, (ii) industrial materials, (iii) energy, and (iv) agricultural products. This broad coverage enables a robust assessment of cross-market return predictability across different commodity groups.

Table 8 compares the performance of image-based return predictions in the Chinese commodity futures market using two approaches: (i) retraining the CNN model on Chinese data and (ii) directly transferring the US-trained CNN model without modification, while also reporting the performance differential ("Transfer – Retrain"). The results reveal systematic underperformance of direct transfer compared with retraining, particularly in capturing cross-sectional return patterns and generating risk-adjusted returns.

Panel A examines short-horizon portfolio performance, showing that whereas direct transfer slightly outperforms retraining in I5/R5 ("H-L" return difference = 0.07%, SR difference = 0.01), this advantage diminishes as the prediction horizon extends. In I20/R5 and I60/R5, Transfer – Retrain "H-L" return differences drop to −0.84% and −1.32%, with SRs of −0.09 and −0.12, respectively. Additionally, direct transfer results in higher turnover, increasing trading frequency without improving returns, reinforcing its failure in short-horizon predictions for Chinese commodity futures.

In the middle-horizon setting (Panel B), Transfer – Retrain shows systematically positive "H-L" return differences, particularly in I20/R20 ("H-L" return = 2.34%, SR = 0.22). However, this improvement stems from retraining underperformance rather than the superiority of direct transfer. The direct transfer models fail to generate consistently strong return differentials, as all Ix/R20 portfolios show "H-L" returns below 1% with near-zero SRs. The lack of stable outperformance suggests direct transfer offers no meaningful advantage over retraining in the Chinese market. Weak results from both methods indicate potential fundamental limitations of image-based models for medium-term return prediction in commodities.

Panel C confirms direct transfer's inapplicability in long-horizon settings. Like the middle-horizon results, although "Transfer – Retrain" reaches 2.58% (SR = 0.36) in I5/R60, this gain results from retraining failures rather than direct transfer effectiveness. As the horizon extends, direct transfer weakens further, with I60/R60 showing negative or zero Transfer – Retrain "H-L" return differences (−0.66%, SR = −0.06). These findings show that direct transfer lacks stable predictive advantages and is particularly ineffective over longer horizons.

In summary, direct transfer fails to systematically outperform retraining and, in most cases, results in weaker return predictability. Although isolated instances of marginal improvement exist, they are neither robust nor consistent across horizons, underscoring the necessity of market-specific adaptation in commodity futures return prediction. The observed differences across these implementation strategies suggest that whereas image-based models have demonstrated success in cross-market transferability within many international stock markets (Jiang et al. 2023), their effectiveness does not extend to commodity futures markets.

We replicate the results of transfer learning using 15 Chinese commodity futures contracts that most closely match the product specifications of their corresponding US counterparts. Table C1 in Appendix C presents these 15 comparable commodity futures and reports the return correlations between their counterparts in the US and Chinese markets. The results indicate generally low correlations—below 0.3 in most cases—except for three metals: nickel (0.41), zinc (0.38), and silver (0.31). As shown in Table C2, the findings are qualitatively consistent with those in Table 8, which uses a broader set of 36 Chinese futures contracts: there is no observed benefit from transfer learning.

Two factors explain why transfer learning does not work in commodity markets. First, commodity futures returns are heavily influenced by localized supply-demand imbalances, government policies, and inventory cycles, which exhibit substantial market-specific variations. Unlike stock markets, where global macroeconomic forces and common risk factors create structural similarities, commodity markets are shaped by idiosyncratic economic and geopolitical factors that limit the generalizability of a model trained in one country to another. Second, Chinese commodity futures markets exhibit distinct characteristics compared with US commodity markets, including differences in liquidity conditions, regulatory structures, and market participant composition, which can significantly impact price formation dynamics.

Recent studies provide further insights into the failure of transfer learning in this context. Chinese commodity futures markets are dominated by short-term retail speculators. As of 2016, individual investors accounted for over 86% of open interest and had an average holding period of less than 4 h, whereas US exchanges are predominantly institution driven (Fan and Zhang 2020). Retail dominance in China interacts with stringent trading rules—such as multitier daily price limits (± 4% for nondelivered contracts and ± 6% for delivery contracts) and strict margin-based position caps that prevent individuals from participating in the delivery month. These constraints shift liquidity away from the front-month contract, and individual traders are restricted to a maximum of 10% of outstanding open interest (Bianchi et al. 2021). Consequently, the market is highly speculative. Aggregate trading volume is 261 times open interest—far above the North American ratio of

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TABLE 8 | Portfolio performance of the CNN model in Chinese commodity futures market.

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R5		I20/R5		I60/R5		I5/R5		I20/R5		I60/R5		I5/R5		I20/R5		I60/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel A: Short-horizon (one-week) portfolio performance																		
Low	2.65	0.26	3.51	0.36	2.38	0.25	2.6	0.27	3.62	0.37	3.14	0.32	−0.08	−0.01	0.17	0.03	1.15	0.18
2	3.41	0.34	3.19	0.32	4.06	0.41	3.76	0.36	3.56	0.36	3.21	0.32	0.53	0.08	0.55	0.09	−1.28	−0.19
High	4.07	0.37	3.74	0.33	3.86	0.33	4.06	0.36	3.3	0.28	3.75	0.32	−0.01	0	−0.67	−0.11	−0.17	−0.02
H-L	1.42	0.2	0.23	0.03	1.48	0.19	1.46	0.2	−0.32	−0.04	0.61	0.08	0.07	0.01	−0.84	−0.09	−1.32	−0.12
Turnover (%)	2181.44		2017.55		1866.16		2196.38		2137.11		2063.71							

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R20		I20/R20		I60/R20		I5/R20		I20/R20		I60/R20		I5/R20		I20/R20		I60/R20	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel B: Middle-horizon (one-month) portfolio performance																		
Low	4.02	0.4	4.88	0.49	4	0.43	3.78	0.39	3.37	0.36	2.84	0.29	−0.36	−0.06	−2.28	−0.36	−1.75	−0.28
2	2.53	0.26	1.5	0.14	1.73	0.17	2.66	0.26	3.27	0.32	4.05	0.4	0.2	0.03	2.66	0.39	3.5	0.52
High	3.28	0.29	3.43	0.31	3.62	0.31	4.07	0.35	3.47	0.29	3.51	0.3	1.18	0.19	0.06	0.01	−0.16	−0.02
H-L	−0.73	−0.09	−1.46	−0.18	−0.38	−0.05	0.29	0.04	0.09	0.01	0.67	0.09	1.54	0.16	2.34	0.22	1.58	0.15
Turnover (%)	517.65		521.32		473.99		509.59		508.75		493.51							

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R60		I20/R60		I60/R60		I5/R60		I20/R60		I60/R60		I5/R60		I20/R60		I60/R60	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel C: Long-horizon (one-quarter) portfolio performance																		
Low	3.66	0.38	2.83	0.31	2.99	0.31	4.16	0.44	2.41	0.25	3.96	0.42	0.76	0.16	−0.64	−0.1	1.47	0.24
2	3.42	0.34	1.83	0.19	2.46	0.26	2.06	0.21	3.22	0.33	2.41	0.25	−2.05	−0.33	2.1	0.31	−0.08	−0.01
High	1.61	0.14	4.11	0.34	4.05	0.34	3.82	0.32	4.79	0.4	4.59	0.37	3.35	0.58	1.03	0.17	0.81	0.11
H-L	−2.05	−0.25	1.28	0.16	1.06	0.13	−0.34	−0.04	2.39	0.3	0.62	0.07	2.58	0.36	1.67	0.17	−0.66	−0.06
Turnover (%)	173.06		164.93		161.26		171.08		163.97		164.88							

Note: This table presents the annualized out-of-sample tercile portfolio performance of the CNN model alongside other strategies in the Chinese commodity futures market. Panels A–C correspond to short-horizon (one-week), middle-horizon (one-month), and long-horizon (one-quarter) performance, respectively. "Ret" is expressed as a percentage (%). Following Jiang et al. (2023), we compare two image-based strategies: one that retrains each CNN model using Chinese commodity futures data and another that directly transfers CNN models trained on US data without retraining.

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31 reported by Fan et al. (2022). In addition, liquidity, volatility, and basis dynamics are concentrated in deferred-month contracts rather than in the nearby contracts that underpin most US price-image studies.

These institutional characteristics create a significant distributional shift for any CNN trained on US data. Daily price limits truncate extreme returns and generate steplike intraday price charts. Meanwhile, liquidity migration to deferred contracts disrupts the consistent front-month term-structure patterns that CNNs interpret as high-order spatial features. The prevalence of retail order flow introduces fat-tailed noise and herding dynamics. Furthermore, global information is primarily absorbed via overnight price gaps, with intraday returns driven largely by domestic order flow, not by US market co-movements (Fung et al. 2013). Taken together, the truncation of tails, liquidity shifts across maturities, and jump-driven information flows violate the covariate stability assumption that underpins transfer learning. The pixel-level distributions, temporal dependencies, and label dynamics observed in Chinese markets diverge substantially from the US data used during model training, thereby preventing effective generalization.

6 | Conclusion

Building on the success of image-based methods in stock market prediction, we extend deep learning applications to commodity futures markets by employing CNNs to analyze OHLC price charts. We bypass the limitations of parametric models, which rely on restrictive assumptions, and instead capture nonlinear price dynamics such as volatility clustering and leverage effects. Empirical results show that CNNs improve return prediction accuracy, particularly over short- to medium-term horizons. Among tested configurations, models trained on 20-day OHLC charts perform most consistently, balancing signal persistence and noise reduction. Portfolios based on CNN predictions achieve superior risk-adjusted returns, as evidenced by higher SRs, highlighting the framework's utility for trading strategies.

We show the CNN models trained on US commodity futures fail on Chinese commodity futures. Structural differences, including supply-demand dynamics and regulations, hinder transferability, highlighting the need for market-specific adaptation. Overall, our work underscores the potential of integrating computer vision with financial econometrics, offering both theoretical insights into price formation and practical tools for market participants.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Futures data were obtained from Commodity Systems Inc. (CSI).

Endnotes

[1] Gao et al. (2025) demonstrate the important role of ChatGPT in forecasting commodity market dynamics. We leave the exploration of how ChatGPT can enhance our results for future research.

[2] For a 20-day window, a third convolutional layer is added, with dilation adjusted to 2×1 and stride to 3×1. For a 60-day window, a fourth convolutional layer is added, with dilation set to 3×1.

[3] To mitigate overfitting, dropout is applied after the fully connected layers, randomly masking 50% of the neurons' outputs to improve the model's generalization. During training, weights are updated using stochastic gradient descent combined with the Adam optimizer (Kingma and Ba 2014; Srivastava et al. 2014). The initial learning rate is set to 1×10^{-5}.

[4] Note that return information appears only in the output target and not as an additional input channel.

[5] Following Han et al. (2016), the TREND factor is computed each month by ranking commodities on composite moving-average trend signals and taking an equal-weighted long position in the top group against an offsetting short position in the bottom group, with the resulting long-minus-short return defining the factor.

[6] In Bianchi et al. (2015), Mom_{12-1} refers to a momentum strategy that, at each month T, ranks commodities into terciles based on their past 12-month returns, going long the top tercile and short the bottom tercile, with a holding period of 1 month. The double-sort strategies, denoted as Mom_{12}–$\mathrm{Ctr}_{\{T\}}$, apply a two-stage procedure: the first sort ranks using 12-month momentum (Mom_{12}), and the second sort applies a contrarian ranking over T months, where T is 18, 24, 36, 48, or 60 months. The double-sort strategy takes long positions in medium-term winners that are long-term losers and short positions in medium-term losers that are long-term winners. All strategies use a 1-month holding period.

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Appendix A

One-Dimensional Models as Benchmarks for Return Prediction

In Appendix A, we use the CNN 1D model and the LSTM model to represent the 1D models serving as benchmarks, allowing us to evaluate the performance of our proposed CNN 2D model against widely used deep learning approaches. Specifically, the CNN 1D serves as a benchmark for models that capture local temporal patterns through convolution along the time axis, while the LSTM serves as a benchmark for models designed to capture nonlinear long- and short-term dependencies in sequential data. This combination allows us to compare the CNN 2D not only against models that focus on local structures but also against those that are designed for longer range temporal dynamics. We conduct the same "Ix/Ry" portfolio analysis as in Table 2, and the corresponding results are reported in Tables A1 and A2.

In the CNN 1D model, the data are represented as a matrix with rows corresponding to time and columns to variables. Convolutional filters in this setting span the full width of the data matrix and slide only along the time dimension. The CNN 1D, as a time-series CNN, processes continuous numerical representations of market data, such as OHLC time series, by applying convolution exclusively along the temporal axis (Jiang et al. 2023). This architecture enables the model to capture local temporal dependencies and short-term patterns but limits its ability to extract the spatial structures embedded in price-chart images. As shown in Table A1, although the CNN 1D achieves positive "H-L" returns in

TABLE A1 | Portfolio performance of the CNN 1D model.

	I5/R5		I20/R5		I60/R5	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel A: Short-horizon (one-week) portfolio performance						
Low	4.22	0.26	3.48	0.21	3.54	0.23
2	4.96	0.29	4.64	0.27	3.63	0.21
High	3.74	0.22	4.75	0.27	5.72	0.32
H-L	−0.48	−0.03	1.27	0.07	2.18	0.13
Turnover (%)	3123.70		2267.89		2021.45	
	I5/R20		**I20/R20**		**I60/R20**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel B: Middle-horizon (one-month) portfolio performance						
Low	4.06	0.25	3.15	0.21	1.07	0.07
2	4.27	0.25	2.57	0.15	5.86	0.35
High	3.73	0.22	6.76	0.38	5.58	0.30
H-L	−0.33	−0.02	3.61	0.20	4.51	0.25
Turnover (%)	726.52		700.85		614.05	
	I5/R60		**I20/R60**		**I60/R60**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel C: Long-horizon (one-quarter) portfolio performance						
Low	5.42	0.34	3.22	0.22	2.09	0.14
2	4.62	0.28	4.63	0.28	4.07	0.24
High	2.96	0.16	5.23	0.26	6.84	0.36
H-L	−2.46	−0.14	2.01	0.11	4.76	0.27
Turnover (%)	244.39		241.80		220.37	

Note: This table reports the portfolio performance of the CNN 1D model in the US commodity futures market. Panels A–C present the results for short-horizon (one-week), medium-horizon (one-month), and long-horizon (one-quarter) performance, respectively. "Ret" is expressed as a percentage (%). The CNN 1D model applies convolutional filters along a single dimension, typically the time axis, capturing local temporal patterns in time-series data without considering spatial relationships across variables. The model uses the same hyperparameter settings (e.g., number of filters, learning rate, early stopping criteria, and kernel stride) as the standard CNN 2D configuration, though the overall parameter count is reduced due to the lower dimensionality of the input data.

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TABLE A2 | Portfolio performance of the LSTM model.

	I5/R5		I20/R5		I60/R5	
	Ret	SR	Ret	SR	Ret	SR
Panel A: Short-horizon (one-week) portfolio performance						
Low	3.09	0.12	3.61	0.09	4.01	0.24
2	6.71	0.39	5.74	0.33	3.79	0.22
High	3.93	0.24	4.96	0.29	4.82	0.27
H-L	0.84	0.10	1.35	0.18	0.80	0.04
Turnover (%)	3105.14		1738.13		2381.43	
	I5/R20		**I20/R20**		**I60/R20**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel B: Middle-horizon (one-month) portfolio performance						
Low	5.26	0.32	1.70	0.04	3.93	0.11
2	0.65	0.04	5.13	0.30	5.76	0.28
High	4.92	0.34	6.40	0.37	5.95	0.33
H-L	−0.34	−0.04	4.70	0.30	2.02	0.20
Turnover (%)	749.46		720.93		516.80	
	I5/R60		**I20/R60**		**I60/R60**	
	Ret	**SR**	**Ret**	**SR**	**Ret**	**SR**
Panel C: Long-horizon (one-quarter) portfolio performance						
Low	6.85	0.43	2.79	0.17	3.00	0.19
2	0.90	0.05	6.94	0.42	6.29	0.38
High	5.40	0.29	3.39	0.18	3.87	0.21
H-L	−1.45	−0.08	0.60	0.03	0.88	0.05
Turnover (%)	240.30		254.36		232.87	

Note: This table reports the portfolio performance of the LSTM model in the US commodity futures market. Panels A–C present results for short-horizon (one-week), middle-horizon (one-month), and long-horizon (one-quarter) performance, respectively. "Ret" is expressed as a percentage (%). The LSTM model, designed to capture both short- and long-term dependencies in sequential data through gated memory cells, is configured with 12 hidden units and a single layer. All other settings, including the learning rate and early stopping criteria, are identical to those used for the CNNs model.

the short-horizon portfolios (e.g., −0.48% with an SR of −0.03 under I5/R5), its performance deteriorates over longer horizons, with several negative or near-zero "H-L" returns, such as −0.33% (SR = −0.02) under I5/R20 and −2.46% (SR = −0.14) under I5/R60.

In contrast, the LSTM model is a recurrent neural network specifically designed to capture both short- and long-term nonlinear dependencies in sequential data. Using gated memory cells, it retains and updates relevant information over time, making it well suited for time-series forecasting tasks (Medvedev and Wang 2022; Chen et al. 2024). However, like the CNN 1D, the LSTM operates strictly along the time dimension and thus cannot exploit the 2D spatial patterns present in chart-based representations. As reported in Table A2, the LSTM shows limited and inconsistent predictive power, with short-horizon "H-L" returns of only 0.84% (SR = 0.10) under I5/R5, and negative or weakly positive "H-L" returns at longer horizons, including −1.45% (SR = −0.08) under I5/R60 and 0.60% (SR = 0.03) under I20/R60.

Overall, these results highlight the consistent advantage of the proposed CNN 2D model over the 1D benchmarks. Unlike the CNN 1D and LSTM, which struggle to deliver stable predictive performance across horizons, the CNN 2D leverages spatial information in price-chart images to achieve superior and more robust portfolio outcomes, as evidenced by the substantially higher "H-L" returns and SRs reported in Table 2.

Appendix B

Logistic Regression Approximation of CNN Predictive Performance

In this appendix, we employ logistic regression to estimate the predictive performance of the CNN model using the underlying market data embedded in the image representations. Following Jiang et al. (2023), the logistic regression specification is similar to Equation (5) and (6), where the dependent variable is the out-of-sample prediction generated by the CNN model on 5-day images (coded as 1 if the predicted probability exceeds 0.5, and 0 otherwise), and the independent variables are the underlying 5-day market data rescaled to mirror the image representations.

Table B1 presents the results of logistic regressions examining the association between image-based forecasts, historical price and liquidity-related factors, and future return realizations across different predictive horizons. The analysis is divided into two primary components. The first set of regressions evaluates the extent to which CNN forecasts are systematically related to lagged price dynamics, trading volume, and trend measures. The second set models the probability of a positive return, incorporating CNN forecasts alongside conventional predictors to assess their relative contributions to return predictability.

The first set of regressions suggests that image-based predictions are significantly influenced by past market behavior, particularly extreme price fluctuations. Notably, prior highs exhibit a strong positive

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TABLE B1 | Logistic regressions using market data with image scaling.

	CNN as Dep. Var.			Positive return indicator as Dep. Var.								
				I5/R5			I5/R20			I5/R60		
	I5/R5 (1)	I5/R20 (2)	I5/R60 (3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
CNN				0.24***		0.23***	0.55***		0.55***	0.90***		0.89***
open_lag_1	0.76	−0.43	−0.59		−7.98***	−8.07***		−8.67***	−8.74***		−9.28***	−9.44***
open_lag_2	−0.51	−2.30**	−1.01		−1.78	−1.75		−1.88	−1.59		−1.53	−1.39
open_lag_3	−0.51	−0.77	0.17		−1	−0.96		−1	−0.86		−1.06	−1.06
open_lag_4	1.03	−0.51	0.83		0.64	0.53		0.58	0.72		0.52	0.42
open_lag_5	−1.22	−0.1	0.42		−0.21	−0.15		−0.34	−0.33		−0.3	−0.42
high_lag_1	−1.27	−0.29	−1.51		−2.57**	−2.47**		−3.28***	−3.23**		−3.33***	−3.13**
high_lag_2	−0.22	0.88	1.26		1.04	1.03		1.18	1.09		1.02	0.78
high_lag_3	−1.16	0.45	−1.03		0.44	0.52		0.55	0.37		0.44	0.69
high_lag_4	0.05	0.19	1.96*		2.04*	2.05*		2.13*	2.11*		2.06*	1.69
high_lag_5	1.58*	1.16	1.15		7.62***	7.53***		7.82***	7.77***		7.64***	7.64***
low_lag_1	−1.77*	0.23	−0.15		−3.67***	−3.57***		−3.87***	−3.90***		−4.13***	−4.23***
low_lag_2	1.16	−0.03	−1.04		0.55	0.46		0.63	0.57		0.44	0.56
low_lag_3	1.55	0.2	−1.59		1.15	1.05		1.05	1.02		0.92	1.21
low_lag_4	0.67	0.19	1.5		1.01	1.04		1.11	1.13		1	0.78
low_lag_5	0.47	1.33	2.24**		9.84***	9.83***		9.87***	9.74***		9.85***	9.75***
close_lag_1	2.80**	2.10*	1.71		1.25	1.12		0.95	0.65		0.76	0.54
close_lag_2	−0.03	0.81	0.36		−1.53	−1.52		−1.81	−1.88		−1.56	−1.56
close_lag_3	−0.87	−1.02	0.35		−0.27	−0.23		−0.3	−0.15		−0.09	−0.32
close_lag_4	0.26	−0.37	−2.57**		−1.23	−1.25		−1.51	−1.49		−1.45	−0.98
ma_lag_1	−1.51*	−0.76	−3.36*		−5.03***	−4.99***		−2.23*	−2.23*		−0.77	−0.18
ma_lag_2	1.26	−3.71**	2.38		0.19	0.17		−1.61	−1.04		−1.75	−2.18
ma_lag_3	−0.56	2.51	−4.46*		0.09	0.1		0.48	0.15		−1.14	−0.38
ma_lag_4	−0.87	−1.61	3.99*		−0.82	−0.78		−0.96	−0.75		2.74	1.97
ma_lag_5	−1.13	1.8	−1.11		0.23	0.32		1.1	0.86		−1.01	−0.73
volume_lag_1	−0.07	0.05	0.07		−0.17***	−0.17***		−0.17***	−0.18***		−0.17***	−0.19***
volume_lag_2	0.15**	−0.03	−0.1		−0.16**	−0.16**		−0.15**	−0.15**		−0.15**	−0.14*
volume_lag_3	−0.1	−0.02	0.09		−0.1	−0.09		−0.1	−0.1		−0.11	−0.14*
volume_lag_4	0.08	0	−0.01		0.16**	0.16**		0.16**	0.16**		0.16**	0.17**
volume_lag_5	−0.04	0.02	−0.03		0.23***	0.23***		0.24***	0.24***		0.24***	0.26***
McFadden R^2	−0.17	−0.17	0.17									
OOS McFadden R^2				−0.1	2.57	2.37	−1.12	2.66	1.46	−2.91	2.65	−0.26

Note: This table presents the results of logistic regressions on out-of-sample CNN forecasts and realized future return indicators based on daily price and volume data. Panel regressions are estimated during the test sample using CNN models trained in the training sample. To compare predictive performance, we report out-of-sample McFadden R^2 values, which assess the cross-entropy loss of the trained model against a benchmark model that uses the future-level in-sample mean as the predictor. Since *ma_lag_1* is defined as the simple average of *close_lag_1* to *close_lag_5*, we exclude *close_lag_5* from the regression to avoid multicollinearity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively, based on Newey–West standard errors. McFadden R^2 values are computed using test sample data.

relationship with CNN forecasts, as seen in *high_lag_5* (1.58, significant at the 10% level in I5/R5) and *high_lag_4* (1.96, significant at the 10% level in I5/R60), indicating that CNN models capture short-term momentum effects, where past peaks signal upward persistence. Conversely, *low_lag_1* (−1.77, significant at the 10% level in I5/R5) suggests that CNN forecasts also incorporate mean-reversion tendencies, with previous downward price movements being associated with lower expected future returns. Additionally, the moving average lag (*ma_lag_1*) shows a significant negative relationship with CNN forecasts (−1.51 in I5/R5 and −3.36 in I5/R60, both at the 10% level), reinforcing the notion that CNN models adjust to deviations from historical smoothed trends rather than relying solely on absolute price levels. Although recent closing prices (*close_lag_1* = 2.80, significant at the 5% level in I5/R5) contribute to short-term forecasts, their impact diminishes over longer horizons, suggesting that CNN models prioritize more structural market signals over time.

The second set of regressions, which models the probability of a positive return, underscores the predictive strength of image-based forecasts. Across all horizons, CNN coefficients remain highly significant,

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confirming that CNN models extract meaningful predictive signals beyond traditional financial indicators. The magnitude of the CNN coefficient increases as the forecast horizon extends (0.24 in I5/R5 to 0.90 in I5/R60), suggesting that deep learning models capture return-relevant patterns that persist over longer time frames. Additionally, prior opening prices exhibit a consistently negative and highly significant relationship with future returns (−7.98 to −9.44, all at the 1% level), reinforcing a mean-reversion effect where higher past opening prices reduce the likelihood of future gains. Meanwhile, prior price highs remain positively associated with future return probabilities (*high_lag_5* = 7.64, significant at the 1% level in I5/R60), consistent with short-term momentum effects. In contrast, prior low prices (*low_lag_1* to *low_lag_5*) show significant negative coefficients (−3.67 to −4.23, all at the 1% level), further supporting the role of mean-reversion tendencies in return dynamics.

The regression results also highlight the influence of trading volume on return predictability, with *volume_lag_1* to *volume_lag_5* being mostly negative and highly significant (−0.17 to −0.19, all at the 1% level), suggesting that higher past trading activity reduces the probability of a positive return. This finding aligns with liquidity-driven price adjustments, where surges in trading volume correspond to price corrections rather than sustained directional trends.

To evaluate model fit, Table B1 reports McFadden R^2 and out-of-sample (OOS) McFadden R^2 values. While McFadden R^2 values remain low across all specifications (−0.17 to 2.66), consistent with the inherent noise in financial return predictions, the OOS McFadden R^2 values indicate that CNN models improve predictive performance relative to traditional benchmarks (2.37 in I5/R5, 1.46 in I5/R20). These findings suggest that although CNN models do not fully explain return variations, they provide incremental predictive power that becomes increasingly relevant at longer time scales. In particular, the logistic regression analysis reveals that CNN models consistently integrate extreme price movements, trend deviations, and liquidity conditions into return predictions.

Appendix C

Replication of Transfer Learning Using Matched Commodity Futures

In this Appendix, we replicate the results of transfer learning using 15 Chinese commodity futures contracts that most closely match the product specifications of their corresponding US counterparts.

TABLE C1 | Return correlations between commodity futures in the US and Chinese markets.

Sector	Commodity	Start date	Correlation
Metals	Aluminum	1999-01	0.259185
	Nickel	2015-04	0.410403
	Lead	2011-04	0.288524
	Zinc	2007-04	0.381278
	Gold	2008-01	0.283164
	Silver	2012-05	0.311421
	Copper	1999-01	0.290257
Agriculture	Soybeans	1999-01	0.114598
	Corn	2004-09	0.124255
	Wheat	2013-01	−0.009913
	Kansas City HRW Wheat[a]	1999-01	0.028520
	Cotton	2004-06	0.273992
	Sugar	2006-01	0.115446
Energy	Low-Sulphur Gasoil	2020-07	0.283205
Livestock	Lean Hogs	2021-01	0.077227

Note: This table presents the 15 comparable commodity futures and reports the return correlations between their counterparts in the US and Chinese markets, adjusted for contract misalignment.
[a]For Kansas City HRW Wheat, we selected hard wheat from the Chinese futures market as the comparable contract, as the two are highly similar in both type and characteristics.

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TABLE C2 | Portfolio performance of the CNN model in Chinese commodity futures market.

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R5		I20/R5		I60/R5		I5/R5		I20/R5		I60/R5		I5/R5		I20/R5		I60/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel A: Short-horizon (one-week) portfolio performance																		
Low	−0.43	−0.05	2.13	0.24	1.6	0.18	4.03	0.45	3.05	0.33	0.68	0.08	6.72	0.79	1.38	0.16	−1.39	−0.16
2	4.34	0.49	2.06	0.24	4.08	0.47	0.38	0.04	2.13	0.25	3.18	0.36	−5.97	−0.68	0.11	0.01	−1.36	−0.15
High	4.38	0.4	3.64	0.32	3.77	0.33	4.22	0.36	2.58	0.22	5.28	0.47	−0.24	−0.03	−1.6	−0.19	2.27	0.24
H-L	4.82	0.43	1.5	0.14	2.17	0.2	0.2	0.02	−0.47	−0.04	4.6	0.42	−6.96	−0.47	−2.98	−0.22	3.66	0.25
Turnover (%)	2033.28		1939.11		1826.65		1990.83		1945.28		1863.42							

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R20		I20/R20		I60/R20		I5/R20		I20/R20		I60/R20		I5/R20		I20/R20		I60/R20	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel B: Middle-horizon (one-month) portfolio performance																		
Low	3.08	0.35	1.85	0.22	1.7	0.2	1.3	0.14	2.14	0.24	2.86	0.32	−2.68	−0.24	0.45	0.06	1.75	0.18
2	3.44	0.39	2.78	0.34	1.66	0.19	4.54	0.51	2.13	0.24	2.43	0.27	1.66	0.17	−0.99	−0.11	1.16	0.12
High	1.55	0.13	3.57	0.3	5.36	0.46	2.64	0.23	3.98	0.35	3.35	0.3	1.65	0.15	0.62	0.06	−3.04	−0.27
H-L	−1.53	−0.13	1.72	0.15	3.67	0.33	1.34	0.11	1.84	0.16	0.49	0.04	4.33	0.23	0.17	0.01	−4.79	−0.28
Turnover (%)	485.92		472.93		464.87		459.99		441.9		442.37							

	Retrain						Direct transfer						Transfer–Retrain					
	I5/R60		I20/R60		I60/R60		I5/R60		I20/R60		I60/R60		I5/R60		I20/R60		I60/R60	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Panel C: Long-horizon (one-quarter) portfolio performance																		
Low	1.33	0.15	1.37	0.17	1.18	0.14	2.14	0.26	2.26	0.26	1.98	0.23	1.22	0.15	1.34	0.15	1.21	0.15
2	3.34	0.38	2.6	0.3	1.96	0.25	2.31	0.27	1.86	0.22	2.2	0.25	−1.57	−0.18	−1.1	−0.11	0.37	0.04
High	2.81	0.25	3.38	0.3	5.6	0.47	4.55	0.36	4.74	0.4	6.02	0.49	2.62	0.29	2.04	0.22	0.63	0.06
H-L	1.48	0.13	2.01	0.19	4.43	0.38	2.41	0.2	2.48	0.22	4.04	0.34	1.4	0.11	0.71	0.05	−0.58	−0.04
Turnover (%)	162.31		162.28		142.84		155.08		148.79		148.45							

Note: This table presents the annualized out-of-sample tercile portfolio performance of the CNN model alongside other strategies in the Chinese commodity futures market with the same commodity futures as Table C1. Panels A–C correspond to short-horizon (one-week), middle-horizon (one-month), and long-horizon (one-quarter) performance, respectively. "Ret" is expressed as a percentage (%). Following Jiang et al. (2023), we compare two image-based strategies: one that retrains each CNN model using Chinese commodity futures data and another that directly transfers CNN models trained on US data without retraining.

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